

Riemann Roch Generalized

Consider the Gauss-Bonnet theorem ([this](#)):

$$\int_M K dA = \int_{\partial M} k_g ds = 2\pi\chi(M)$$

where M is a 2-dimensional compact Riemann manifold, K the Gaussian curvature, k_g the geodesic curvature, and $\chi(M)$ the Euler characteristic. This can be thought of as a special case of The Riemann-Roch Theorem!

There is also a more direct topological generalization of the Gauss-Bonnet theorem known as the Chern theorem, or Chern–Gauss–Bonnet theorem, to $2n$ -dimensional Riemann manifolds without boundary:

$$\chi(M) = \int_M e(\Omega)$$

where $\chi(M)$ is the Euler characteristic, Ω is the curvature form (or Levi-Civita connection), and $e(\Omega)$ is the Euler class:

$$e(\Omega) = \frac{1}{(2\pi)^n} \text{Pf}(\Omega)$$

where Pf is the *Pfaffian*; namely as Ω can always be represented as a skew-symmetric matrix (when the dimension of M is even, hence the $2n$) it can always be written as the square of a polynomial in matrix entries, namely if A is the skew-symmetric matrix:

$$\text{Pf}(A)^2 = \det(A)$$

for example:

$$A = \begin{pmatrix} 0 & a \\ -a & b \end{pmatrix} \quad \text{Pf}(A) = a$$

$$B = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix} \quad \text{Pf}(B) = af - be + dc$$

Atiyah–Singer index theorem seems to generalize what I’m working on further.

Rapid fire concepts we know:

- We shall take varieties and schemes for granted
- We shall be working with algebraic curves, usually algebraic plane curves; hence curves given by $f(x, y)$ for coordinate functions x, y .
- Such curves are projective ([1]). If they are smooth, we naturally ask about their relation to one-dimension complex manifolds; these are called Riemann surfaces (as they are 2 real dimensional)

- Compact Riemann surfaces are classified up to homeomorphism by their genus, and $\chi(X) = 2 - 2g$ where χ is the Euler Characteristic.
- Via normalization, we have the following (This is in fact proven in [3])

Theorem 0.1: Characterizing Riemann Surfaces Using Normalization

Any compact Riemann surface $\tilde{C}$ can be obtained through the normalization of a certain plane algebraic curve C with at most ordinary double points, i.e. There exists a holomorphic mapping $\sigma : \tilde{C} \rightarrow \mathbb{P}^2(\mathbb{C})$ such that $\sigma(\tilde{C})$ is an algebraic curve possessing at most ordinary double points

- The geometry polynomial function (by FTC) makes them surjective and n -to-1 except at the ramification points.
- We shall take holomorphic and meromorphic differential forms for granted
- Key result about holomorphic differential forms:

Theorem 0.2: Stokes Theorem For Holomorphic Differentials

Let C be a Riemann surface, $\Omega \subseteq C$ an open set, $\overline{\Omega}$ compact and $\partial\Omega = \gamma$ a piecewise smooth curve, and ω a holomorphic differential defined on an open set containing $\overline{\Omega}$. Then:

$$\int_{\partial\Omega} \omega = 0$$

Proof :

use local coordinates and apply Cauchy's theorem to get

$$\int_{\partial\Omega_i} \omega = 0$$

on the components, which then gluing gives the desired result

- Residue is winding number

1 Poincaré-Hopf Formula

Let C be a compact Riemann surface. Then we know that for any nonconstant $f \in K(C)$ (the set of meromorphic functions on C), we get:

$$\sum_{p \in C} \nu_p(f) = 0 \tag{1}$$

that is, the number of zeros plus the number of poles (counting multiplicity) is zero. Phillip actually has this really cool generalization of the proof that I got in [2], he prove the following:

Theorem 1.1: Residue Theorem For Forms

Let C be a compact Riemann surface and $\omega \in K^1(C)$ a differential form (the K^1 is his notation for diff. form). Then:

$$\sum_{p \in C} \text{Res}_p(\omega) = 0$$

Proof :
exercise

Then you can deduce the usual equation (1) by setting $\omega = \frac{df}{f}$. Now, we can ask the follow: for a trivial $\omega \in K^1(C)$, what is:

$$\sum_{p \in C} \nu_p(\omega)$$

A general differential form on a compact Riemann surface has more flexibility than functions, and the order at p is certainly well-defined the same way the residue of a differential form is well defined, so this is a meaningful question. As one-dimensional differential forms are co-vector fields, this is equivalent to a problem on vector-fields, and we know that one interpretation of the Euler-characteristic on surfaces is to count the zero-points (which are sink and source points) and the saddle points (notice how closely this naturally relates to curvature; this is why it comes up naturally in the Gauss-Bonnet Theorem) and so we get:

$$\sum_{p \in C} \nu_p(\omega) = -\chi(C)$$

- There is an interesting word on the schematic representation of curves, [3, p.28]
- Note that meromorphic functions on complex manifold need not be able to map to $\mathbb{CP}^1$, consider z/w from $\mathbb{C}^2$.
- The end of chapter I, section 10, has a very very simple proof on when one can embedd a Riemann surface into $\mathbb{CP}^n$!

2 About Normalization

- He starts with a general plane curve, defines singular points to be when the three partial are zero (hence no linear term when translated to $[0 : 0 : 1]$, or in his notation $[1 : 0 : 0]$). We can then consider any curve $f(x, y) = F(1, x, y)$ when considering the canonical embedding $(x, y) \mapsto [1 : x : y]$. Then it must be double/triple/tuple if the lowest degree is $\geq 2, \geq 3, \geq k$. It is or ordinary if all the tangent line are distinct.
- We can now consider irreducible components. If it is over a UFD, then the GCD is well-defined and so we get that f, g have no common factors if $(f) + (g) = (1)$ so that if $(f) + (g) = (h)$ then $f = Fh$ and $g = Gh$ so that $fG = gF$. We can detect this using the resultant, and hence we can detect singular points over non-characteristic 0 fields using the discriminant. We can then proceed to show that if C is a curve, S the singular points, and $C^* = C \setminus S$ the compliment of the singular points, then C^* is connected, not necessarily compact, S is finite.

- Next comes normalization which he defines as follows:

Definition 2.1: Normalization

Suppose C is a irreducible plane algebraic curve, S the set of its singular points. Then there exists a compact Riemann surface $\tilde{C}$ and a holomorphic mapping

$$\sigma : \tilde{C} \rightarrow \mathbb{CP}^2$$

such that

1. $\sigma(\tilde{C}) = C$
2. $\sigma^{-1}(S)$ is a finite set
3. $\sigma : \tilde{C} \setminus \sigma^{-1}(S) \rightarrow C \setminus S$ is injective

Then $(\tilde{C}, \sigma)$ is the normalization of C , and we write just $\tilde{C}$ if it is unambiguous in context.

- We get the following result equivalent to birational maps:

Lemma 2.2: Dense Injective, then Biholomorphic

Suppose $\tilde{C}, \tilde{C}'$ are Riemann surfaces and

$$h : \tilde{C} \rightarrow \tilde{C}'$$

is a surjective holomorphic mapping which is injective on a dense open subset of $\tilde{C}$. Then h is a biholomorphic mapping.

Proof :

Prove it is biholomorphic in every neighbourhood.

- Then

Theorem 2.3: Uniqueness of Normalization

The normalization of an algebraic curve C is unique up to isomorphism (it commutes in the way you expect)

Proof :

use above lemma.

The goal now is to prove a normalization always exists.

- Next Griffiths defines $\mathbb{C}\{x\}$ which is not quite $\mathbb{C}[[x]]$ because it is the set of all *convergent* power series of the form $\sum_k a_k x^k$. Similarly for $\mathbb{C}\{x, y\}$ which is convergent $\sum_{i,j} a_{ij} x^i y^j$.

Then a Weierstrass polynomial is a polynomial of the form. He denotes the later by $\mathcal{O}$.

$$w = y^d + a_1(x)y^{d-1} + \cdots + a_d(x)$$

where

$$a_j(x) \in \mathbb{C}\{x\} \quad a_j(0) = 0 \quad j = 1, \dots, d$$

Then he goes on to prove the *Weierstrass Propagation theorem* which say that under the appropriate generic conditions, any $f \in \mathcal{O}$ can be written as $f = u \cdot w$ where u is a unit (so $1/u \in \mathcal{O}$ and w is a Weierstrass polynomial.

- Proving this, we get the following great corollary:

Corollary 2.4: Ring Structure of Analytic Func

$\mathcal{O}$ is a UFD, hence any $f \in \mathcal{O}$ can be expressed up to unit u as:

$$f = uf_1f_2 \cdots f_n$$

where each f_i is irreducible in $\mathcal{O}$.

Proof :

Note that proving $\mathbb{C}\{x\}$ is a UFD is easy because any $f(x) = x^m h(x)$ with $h(x)$ a unit with nonzero constant. For the rest checkout [3].

- An important realization is that being irreducible in $\mathbb{C}\{x\}[y]$ and $\mathbb{C}\{x, y\}$ is the same thing.
- Next, let's take an irreducible curve, center the point of interest at $(0, 0)$, represent it as

$$y^n + a_1(x)y^{n-1} + \cdots + a_d(x)$$

Then locally at 0, we may have multiple local irreducible components. Let's take the curve over $\mathbb{C}^2$. Then around 0, if there are multiple components, it will look like multiple disks that are glued together at a point.

- Using this, we can locally normalize, and then use that to globally normalize
- Next Griffith looks at intersection numbers and Bézout Theorem.

Ramification Divisors and the Riemann-Hurwitz Formula

Let $\nu_f(p)$ be the order of a point of a holomorphic mapping $f : C \rightarrow C'$ between Riemann surfaces. As such a map is always surjective if it is non-constant, we have $\nu_f(p) \geq 1$. It ramifies if $\nu_f(p) > 1$, and as ramified points are isolated we know that $\{p \in C \mid \nu_f(p) > 1\}$ is finite. Thus we have the well-defined:

Definition 2.5: Ramification Divisor

The *ramification divisor* of f is :

$$R = \sum_{p \in C} (\nu_f(p) - 1)p \in \text{Div}(C)$$

Recall that if $D = \sum_{p \in C} n_p \cdot p$ is a divisor, $\deg D = \sum_{p \in C} n_p$.

Theorem 2.6: Riemann-Hurwitz Formula

Let C, C' be compact Riemann surfaces where:

$$\text{genus}(C) = g \quad \text{and} \quad \text{genus}(C') = g'$$

Let $f : C \rightarrow C'$ be a nonconstant holomorphic mapping of degree n . Let R be the ramification divisor of f . Then:

$$\deg R = 2(g + n - ng' - 1)$$

Proof :

First, we shall use the fact that on any compact Riemann surface, there exists a nontrivial meromorphic differential. This could be intuitively seen by the correspondence between compact Riemann surfaces and curves, and on any curve we have a non-trivial meromorphic differential.

Hence, choose $\omega \in K^1(C')$, and let $f^*\omega$ be its pullback. Locally we have:

$$\omega = z^\nu$$

and

$$\omega = g(\omega)d\omega$$

Hence:

$$f^*\omega = \nu g(z^\nu) z^{\nu-1} dz$$

and so the order of each point is:

$$\nu_{f^*\omega}(p) = \nu_f(p)\nu_\omega(f(p)) + \nu_f(p) - 1$$

and so the divisor of $(f^*\omega)$ is :

$$\begin{aligned} (f^*\omega) &= \sum_p \nu_f(p)\nu_\omega(f(p))p + R \\ &= \sum_q \nu_\omega(q) \left(\sum_{f(p)=q} \nu_f(p)p \right) + R \\ &= \sum_q \nu_\omega(q) f^{-1}q + R \\ &= f^{-1} \left(\sum_q \nu_\omega(q)q \right) + R = f^{-1}((\omega)) + R \end{aligned}$$

Thus:

$$\deg(f^*\omega) = \deg f^{-1}((\omega)) + \deg R$$

Then from the Poincaré-Hopf formula:

$$\deg(f^*\omega) = 2g - 2 \quad \deg(\omega) = 2g' - 2$$

Thus directly computing we get:

$$\begin{aligned} \deg f^{-1}(\omega) &= \sum_q \nu_\omega(q) \left(\sum_{f(p)=q} \nu_f(p) \right) \\ &= \deg(\omega) \deg f \\ &= (2g' - 2)n \end{aligned}$$

then substituting we get:

$$2g - 2 = (2g' - 2)n + \deg R$$

re-arranging gives our desired result.

From this we can immediately conclude:

Corollary 2.7: Holomorphic Map and Genus

Let $f : C \rightarrow C'$ be a holomorphic map between genus surfaces of genus g, g' . Then if $n \geq 1$:

$$g \geq g'$$

Proof :

Apply the formula.

Corollary 2.8: Compact Riemann Surface As Algebraic Equation

Let C be a compact Riemann surface and let

$$x, y : C \rightarrow \mathbb{CP}^1$$

be holomorphic mapping where $\deg x = d$. Then there exists a polynomial equation f such that $f(x, y) = 0$.

Proof :

[3, p.94]

Next, we cover the genus formula.

Theorem 2.9: Genus Formula

Let $C \subseteq \mathbb{CP}^2$ be an irreducible algebraic curve of degree d with δ singular points all being ordinary double points, and let $\sigma : \tilde{C} \rightarrow C$ be its normalization where $\text{genus}(\tilde{C}) = g$. Then:

$$g = \frac{(d-1)(d-2)}{2} - \delta$$

Proof :
[3, p.95].

3 The Riemann-Roch Theorem

Let C be a compact Riemann surface. Define the divisor to be the formal sum:

$$D = \sum_i^k n_i p_i \in \text{Div}(C)$$

A divisor is called *effective* if each coefficient is ≥ 0 . and we write $D \geq 0$. Any divisor can be expressed as the difference of two effective divisors. We now define the Riemann-Roch space:

$$\mathcal{L}(D) = \{f \in K(C) \mid (f) \geq -D\}$$

That is, $\mathcal{L}(D)$ contains all $f \in K(C)$ wheter f has a pole of multiplicity at most n_i at p_i if $n_i > 0$ and a zero of multiplicity at least $|n_i|$ at p_i if $n_i < 0$. We can then dfine:

$$l(D) = \dim \mathcal{L}(D)$$

We then define a similar notion for divisors:

$$K^1(D) = \{\omega \in K^1(C) \mid (\omega) \geq D\}$$

where now, if $n_i < 0$ then ω can have a p_i as a pole of multiplicity at most $|n_i|$ and if $n_i > 0$ then ω must have a p_i as a zero of multiplicity at least n_i . Again , $K^1(D)$ is a vector space and :

$$\dim K^1(D) = i(D)$$

We shall usually be focusing on effect divisors; $D \geq 0$; in that case we have that $\mathbb{C} \subseteq \mathcal{L}(D)$. Furthermore, to be more align with the usual differential geometry notation, we shall write $\Omega^1(D)$ instead of $K^1(D)$. Then for $D \geq 0$, $\omega \in \Omega^1(D)$ means ω is holomorphic and has a zero of appropriate multiplicity at p_i for appropriate points given by D .

Definition 3.1: Linearly Equivalent

Let $D, E \in \text{Div}(C)$. Then D, E are *linearly equivalent* if there exists a $f \in K(C)$ such that

$$(f) = D - E$$

and we write $D \sim E$.

We can also write $D = (f) + E$ for some (f) , and hence we are really saying that $D \sim E$ is they are equal up to principal divisor. Hence, two divisor are linearly equivalent if they belong to the same coset in $\text{Div}(C)/\text{PDiv}(C)$.

Proposition 3.2: Dimension and Linear Equivlaence

Let $D, E \in \text{Div}(C)$, $D \sim E$. Then:

$$\mathcal{L}(D) \cong \mathcal{L}(E) \quad \Omega^1(D) \cong \Omega^1(E) \quad \deg D = \deg E$$

Proof :

Say $D \sim E$. Then there exists an $f \in K(E)$ such that

$$(f) = D - E$$

Now pick $g \in CL(D)$ so that $(g) + D \geq 0$. Then:

$$(fg) + E = (f) + (g) + E = D - E + (g) + E = (g) + D \geq 0$$

Thus, there is a mapping:

$$\mathcal{L}(D) \rightarrow \mathcal{L}(D) \quad g \mapsto fg$$

This mapping is certainly linear, and it is bijective as the inverse is:

$$\mathcal{L}(E) \rightarrow \mathcal{L}(D) \quad g \mapsto \frac{g}{f}$$

Thus, $\mathbb{C}(D) \cong \mathcal{L}(E)$. Similarly, we certainly get that $\Omega^1(D) \cong \Omega^1(E)$. TO show the degree's are the same, recall that for any $f \in K(C)$:

$$\sum_{p \in C} \nu_p(f) = 0$$

Then:

$$\sum \nu_p(f) = \deg(f) = \deg D - \deg E$$

hence $\deg E = \deg D$, as we sought to show.

Proposition 3.3: Brill-Noether Reciprocity

Let $\omega \in \Omega^1(C)$ and $(\omega) = D + E$. Then:

$$\mathcal{L}(D) \cong \Omega^1(E) \quad \mathcal{L}(E) \cong \Omega^1(D)$$

Proof :

Let $f \in \mathcal{L}(D)$ so that $(f) \geq -D$. Then:

$$(f\omega) = (f) + (\omega) \geq D + D + E = E$$

Thus define:

$$\mathcal{L}(D) \rightarrow \Omega^1(E) \quad f \mapsto f\omega$$

which is certainly a vector-space homomorphism with inverse:

$$\Omega^1(E) \rightarrow \mathcal{L}(D) \quad \varphi \mapsto \frac{\varphi}{\omega}$$

showing they are isomorphic. By symmetry, we have the similar isomorphism by swapping E and D , completing the proof.

Definition 3.4: Laurent Principal Part

Let $\sum_{i=-n}^{\infty} a_i z^i$ be a Laurent series. Then its singular part

$$\sum_{i=-n}^{-1} a_i z^i$$

is the *Laurent principal part*, or just *principal part*. If $a_{-n} \neq 0$, then N is called the order of the Laurent principal part.

Now suppose C is a Riemann surface, D an effective divisor. Let $z_i, i = 1, 2, \dots, k$ be local coordinates near p_i and $z_i(p_i) = 0$. Then if for every p_i , there is given a Laurent principal part of order n_i :

$$\eta_i = \frac{a_{in_i}}{z_i^{n_i}} + \dots + \frac{a_{i1}}{z_i}$$

we can ask the following question: under what condition does there exist an $f \in K(C)$ such that in a neighbourhood of each p_i , f has η_i as a Laurent principal part? This is called the *Mittag-Leffler problem*, and answering it will allow us to deduce the Riemann-Roch Theorem.

First off, a meromorphic function on a compact Riemann surface C is up to a constant uniquely determined by its Laurent principal part, for if $f_1, f_2 \in K(C)$ that have the same Laurent principal part then $f_1 - f_2$ is holomorphic on C , and hence is constant. Thus, we may identify $\mathcal{L}(D)/\mathbb{C}$ as a subspace of $\mathbb{C}^d$, $d = \deg D$, as follows. Fix $D = \sum_i n_i p_i \geq 0$, and let z_i be a coordinate function around p_i where $z_i(p_i) = 0$. Then for $f \in \mathcal{L}(D)$, let η_i be the principal part at p_i . Then we know that:

$$\eta_i = \frac{a_{in_i}}{z_i^{n_i}} + \dots + \frac{a_{i2}}{z_i^2} + \frac{a_{i1}}{z_i}$$

where $a_{in_i}, \dots, a_{i1} \in \mathbb{C}$. Now let P_i be the n_i -dimensional vector space over $\mathbb{C}$ spanned by $\{1/z_i^{n_i}, \dots, 1/z_i^2, 1/z_i\}$. Then we have the injective vector-space homomorphism:

$$\frac{\mathcal{L}(D)}{\mathbb{C}} \rightarrow P_1 \oplus \dots \oplus P_k \quad [f] \mapsto (\eta_1, \dots, \eta_k)$$

Then by identifying each P_i with $\mathbb{C}^{n_i}$, we get that this space is a subspace of $\mathbb{C}^d$. Note that surjectivity is not so obvious or even necessary because we don't know if these maps are compatible (ex. the gluing conditions can impose restrictions).

3.1 Dimension of $\Omega^1(C)$

The goal is to show the following:

Theorem 3.5: Dimension of Differential Forms

Let C be a compact Riemann surface of genus g . Then:

$$\dim \Omega^1(C) = g$$

Note that this is easy in the case where $g = 0$, this is the Riemann sphere and we showed that the only holomorphic differential forms on the Riemann sphere are constant. The proof will proceed in two parts:

Proposition 3.6: Dimension of Differential Form: I

Let C be a compact Riemann surface of genus g . Then:

$$\dim \Omega^1(C) \geq g$$

The key idea is that we can “embed” the space of homogeneous functions of the right degree (minus the singularities) into $\Omega^1(C)$ by essentially taking it’s differential with a bit of normalization.

Proof :

First, by normalization, for a compact Riemann surface C , there exists a holomorphic mapping $\pi : C \rightarrow \mathbb{P}^2$ such that $C' = \pi(C)$ is an irreducible algebraic curve of degree d in $\mathbb{P}^2$ possessing δ ordinary double points, say $p_1, \dots, p_\delta$, and no other singular points. Then if $d = 1$ or $d = 2$, then by the genus formula $g = 0$. Thus we shall assume $d \geq 3$.

Now suppose C is mapped holomorphically onto $\mathbb{P}^2$. We may apply a transformation of coordinates so that C' is not tangent to the line at infinity L_∞ in $\mathbb{P}^2$, that L_∞ does not pass through the points $p_1, \dots, p_\delta$, and that C' does not pass through $[0 : 0 : 1]$ (so the origin, $(0, 0)$). Indeed, these conditions can be guaranteed due to the following. We can guarantee that C' and L_∞ for we can always find a linear transformation so that C' intersects a line at d distinct points. Taking this line to be L_∞ , suppose C' and L_∞ are tangent. Then at the point of tangency, their intersection number would be 2, and so we get by Bézout that $(C' \cdot L_\infty) \geq d + 1$, a contradiction. We can then do a similar reasoning to guarantee L_∞ does not pass through $p_1, \dots, p_\delta$. Finally, for the $[0 : 0 : 1]$ condition, take a point which is on L_∞ but not on C' , and apply a transformation so that the point becomes $[0 : 0 : 1]$. Then we got our desired coordinates.

We can now write out our affine equation for C' such that $f(x, y) = 0$. Let $\Gamma = p_1 + \dots + p_\delta$, and let S^{d-3} represent the set of homogeneous polynomials with complex coefficients of degree $d - 3$ in three variables. Let:

$$S^{d-3}(-\Gamma) = \{G(\zeta^0, \zeta^1, \zeta^2) \in S^{d-3} \mid G(p_i) = 0, \forall i = 1, 2, \dots, \delta\}$$

Note that by some counting arguments we get:

$$\dim S^{d-3} = \frac{(d-1)(d-2)}{2}$$

On the other hand, as the collection of polynomials that vanish at p_i is equivalent to the fact that the coefficients of this polynomial satisfy a linear equation, we get:

$$\dim S^{d-3}(-\Gamma) \geq \frac{(d-1)(d-2)}{2} - \delta \stackrel{!}{=} g$$

where $\stackrel{!}{=}$ comes from the genus formula, theorem 2.9. Thus, if we show there is an injective homomorphism from $S^{d-3}(-\Gamma)$ into $\Omega^1(C)$, we have that:

$$\dim \Omega^1(C) \geq \dim S^{d-3}(-\Gamma) = g$$

To that end, for any $G \in S^{d-3}(-\Gamma)$. We need to find a form $\omega_G \in \Omega^1(C)$ for it to map to. What we will do is find a pullback of a form α along π . Let $g(x, y) = G(1, x, y)$, $U_0 = \{[1, x, y] \in \mathbb{P}^2\}$ and $U_1 = \{[u, 1, v] \in \mathbb{P}^2\}$. Define the following:

$$\alpha_g = \begin{cases} \frac{g(x, y) dx}{\frac{\partial f(x, y)}{\partial y}} \Big|_{C'}, & \text{on } C' \cap U_0, \text{ when } \frac{\partial f(x, y)}{\partial y} \Big|_{C'} \neq 0 \\ -\frac{g(x, y) dy}{\frac{\partial f(x, y)}{\partial x}} \Big|_{C'}, & \text{on } C' \cap U_0 \text{ when } \frac{\partial f(x, y)}{\partial y} \Big|_{C'} = 0 \\ -\frac{g\left(\frac{1}{u}, \frac{v}{u}\right) \frac{dv}{u^2}}{\frac{\partial f}{\partial y}\left(\frac{1}{u}, \frac{v}{u}\right)} \Big|_{C'}, & \text{on } C' \cap U_1 \end{cases}$$

where $\frac{\partial f}{\partial y}(1/u, v/u)$ is the value of $\frac{\partial f}{\partial y}$ at $(1/u, v/u)$. As C' does not pass through the point $[0, 0, 1]$, α is defined over all of C' . It is in fact uniquely defined. Indeed, inside U_0 the affine equation is $f(x, y) = 0$ and thus on C' :

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

and thus:

$$\frac{dx}{\frac{\partial f(x, y)}{\partial y}} \Big|_{C'} = -\frac{dy}{\frac{\partial f(x, y)}{\partial x}} \Big|_{C'}$$

Thus, we have the linear homomorphism:

$$S^{d-3}(-\Gamma) \rightarrow K^1(C) \quad G \mapsto \omega_G = \pi^*(\alpha_g)$$

Note this map is injective since if $\omega_G = 0$, then C' is contained in the curve $g(x, h) = 0$, but this is impossible since $\deg G = d - 3$. We must prove that ω_G is indeed holomorphic, and hence it maps into $\Omega^1(C)$, which shall complete the proof.

To that end, we must show that there are no poles. If a point $p \in C'$ is a pole of ω_G on $\pi^{-1}(C' \cap U_0)$ can become a pole of ω_G , then:

$$\frac{\partial f(x, y)}{\partial y} \Big|_{p'} = \frac{\partial f(x, y)}{\partial x} \Big|_{p'} = 0$$

where $p' = \pi(p)$. Then p' is a double point of C' , so $p' \in \Gamma$, so

$$g(x, y)|_{p'} = 0$$

But p' is a second-order zero of F , and hence a first order zero of $\partial f / \partial x$, so:

$$\alpha|_{p'} = -\frac{g(x, y) dy}{\frac{\partial f(x, y)}{\partial x}} \Big|_{C'} \neq \infty$$

hence ω_G is holomorphic at p .

Now consider ω_G on $\pi^{-1}(C' \cap U_1)$. Then by definition of α :

$$\alpha = -\frac{u^{d-3} g\left(\frac{1}{u}, \frac{v}{u}\right) du}{u^{d-1} \frac{\partial f}{\partial y}\left(\frac{1}{u}, \frac{v}{u}\right)} \Big|_{C'}$$

to simplify notation, denote the denominator as $h(u, v)$. Then we must show $h(u, v) \neq 0$ on $C' \cap L_\infty$. To see this, note that the equation of C' on U_1 is

$$k(u, v) = u^d f\left(\frac{1}{u}, \frac{v}{u}\right) = 0$$

And notice:

$$\frac{\partial k(u, v)}{\partial v} = u^d \frac{\partial f}{\partial u}\left(\frac{1}{u}, \frac{v}{u}\right) \cdot \frac{1}{u} = h(u, v)$$

With this setup, suppose $p' \in C' \cap L_\infty$ such that

$$h(u, v)|_{p'} = 0$$

Then since p' is not a double point of C' , it must be that :

$$\frac{\partial k(u, v)}{\partial u}\bigg|_{p'} \neq 0$$

Since $p' \in L_\infty$, the coordinates of p' must be of the form $[0, 1, v]$. Then we can write the tangent line to C' at p' as :

$$\frac{\partial k(u, v)}{\partial u}\bigg|_{p'}(u - 0) = 0$$

But then the tangent line is simply L_∞ , which contradicts our choice of coordinates, hence no such point exists, and ω_G is holomorphic, as we sought to show.

Let us now show the other direction of $\dim \Omega^1(C) \leq g$. This direction shall require some differential geometry reasoning.

Let C be a compact Riemann surface of genus g with $\gamma_1, \gamma_2, \dots, \gamma_{2g-1}, \gamma_{2g}$ framing a $\mathbb{Z}$ -basis of the homology group $H_1(C, \mathbb{Z})$. Then if γ is a closed curve on C ,

$$\gamma = \sum_{i=1}^{2g} n_i \gamma_i + \partial\Omega$$

where Ω is a region on C . Let $[\gamma]$ represent the homology class containing γ .

Proposition 3.7: Natural Dual of Homology Group

Let γ be a closed differential one-form on C . Then the mapping:

$$\eta_\lambda : H_1(C, \mathbb{Z}) \rightarrow \mathbb{C} \quad [\gamma] \mapsto \int_\gamma \lambda$$

is well-defined

Proof :

The key is that γ is closed. If γ' is homologous to γ then $\gamma - \gamma' = \partial\Omega$ where Ω is a region on

C . By stokes theorem

$$\eta_\lambda(\gamma) - \eta_\lambda(\gamma') = \int_{\gamma - \gamma'} \lambda = \int_{\partial\Omega} \lambda = \iint_{\Omega} d\lambda = 0$$

Proposition 3.8: Exactness

Let λ be a closed differential one-form on C . Then if for all i ,

$$\int_{\gamma_i} \lambda = 0$$

then λ is exact.

Proof :

We must find f such that $df = \lambda$. Let $p_0 \in C$ be some fixed point. Then for any point $p \in C$, choose a path γ from p_0 to p and define:

$$f(p) = \int_{\gamma} \lambda$$

If this is well-defined, this is the function we need. Indeed, if γ' is another path from p_0 to p then $\gamma - \gamma'$ is a closed curve on C , and hence:

$$\gamma - \gamma' = \sum_i^{2g} n_i \gamma_i + \partial\Omega$$

where $n_i \in \mathbb{Z}$ and Ω is a region on C . Then as:

$$\int_{\gamma_i} \lambda = 0 \quad \forall i = 1, \dots, 2g$$

and

$$\int_{\partial\Omega} \lambda = \iint_{\Omega} d\lambda = 0$$

we get:

$$\int_{\gamma} \lambda = \int_{\gamma'} \lambda$$

Thus, f is a well-defined smooth function on C , and by the FTA we get $df = \lambda$, as we sought to show.

Proposition 3.9: Cauchy Riemann For Differential Forms

Let C be a compact Riemann surface, and $\omega, \varphi \in \Omega^1(C)$. Then if:

$$\omega + \bar{\varphi} = df$$

then

$$\omega = \varphi = 0$$

Proof :

Let $\omega = h(z)dz$ and $\varphi = g(z)dz$. Then as C is one-dimensional:

$$\omega \wedge \varphi = 0$$

and

$$\frac{i}{2} \varphi \wedge \bar{\varphi} = |g(z)|^2 \frac{i}{2} dz \wedge d\bar{z} = |g(z)|^2 du \wedge dv$$

We shall show $\varphi = 0$. For the sake of contradiction suppose $\varphi \neq 0$. Then:

$$\frac{i}{2} \iint_C \varphi \wedge \bar{\varphi} = \iint_C |g(z)|^2 du \wedge dv > 0$$

But then:

$$\varphi \wedge \bar{\varphi} = \varphi \wedge \omega + \varphi \wedge \bar{\varphi} = \varphi \wedge (\omega + \bar{\varphi}) = \varphi \wedge df$$

If we show $\iint_C \varphi \wedge df = 0$, we are done. Indeed, notice that:

$$d(f\varphi) = df \wedge \varphi + f \wedge d\varphi$$

As $\varphi \in \Omega^1(C)$, g is holomorphic, and so:

$$\frac{\partial g}{\partial \bar{z}} = 0$$

Thus:

$$\begin{aligned} d\varphi &= d(gdz) = dg \wedge dz \\ &= \left(\frac{\partial g}{\partial z} dz + \frac{\partial g}{\partial \bar{z}} d\bar{z} \right) \wedge dz = 0 \end{aligned}$$

Hence:

$$\varphi \wedge df = -df \wedge \varphi = -d(f\varphi)$$

Note that $f\varphi$ is a differential one-form, so:

$$\iint_C d(f\varphi) = 0$$

But then:

$$\iint_C \varphi \wedge df = - \iint_C d(f\varphi) = 0$$

By a similar argument $\omega = 0$, completing the proof.

Proposition 3.10: Dimension of Differential Form: II

Let C be a compact Riemann surface of genus g . Then:

$$\dim \Omega^1(C) \leq g$$

Proof :

For the sake of contradiction suppose $\dim \Omega^1(C) \geq g + 1$ so that there are $g + 1$ elements $\omega_1, \dots, \omega_{g+1} \in \Omega^1(C)$ that are $\mathbb{C}$ -linearly independent. Take the equations:

$$\int_{\gamma_i} \sum_{j=1}^{g+1} (\lambda^j \omega_j + \eta^j \bar{\omega}_j) = 0 \quad i = 1, 2, \dots, 2g$$

where each γ_i is given by choosing a $\mathbb{Z}$ -basis for $H_1(C, \mathbb{Z})$. By treating $\lambda^1, \dots, \lambda^{g+1}, \eta^1, \dots, \eta^{g+1}$, then the above is $2g$ linear equations with $2g + 2$ unknowns. This system must then have a trivial solution, suppose $\lambda_0^1, \dots, \lambda_0^{g+1}, \eta_0^1, \dots, \eta_0^{g+1}$ is a solution. Then choosing:

$$\omega = \sum_{j=1}^{g+1} \lambda_0^j \omega_j \in \Omega^1(C) \quad \varphi = \sum_{j=1}^{g+1} \eta_0^j \bar{\omega}_j \in \Omega^1(C)$$

then for all $i = 1, 2, \dots, 2g$:

$$k \int_{\gamma_i} (\omega + \bar{\varphi}) = \int_{\gamma_i} \sum_{j=1}^{g+1} (\lambda_0^j \bar{\omega}_j + \eta_0^j \omega_j) = 0$$

This implies it is exact, hence there exists a C^∞ function f such that

$$\omega + \bar{\varphi} = df$$

But then by proposition 3.9 $\omega = \varphi = 0$. But then as $\lambda_0^1, \dots, \lambda_0^{g+1}, \eta_0^1, \dots, \eta_0^{g+1}$ is a non-trivial solution so that they are not all 0, this contradicts the linear independence of $\omega_1, \dots, \omega_{g+1}$, as we sought to show.

Theorem 3.11: Dimension of Differential Forms

Let C be a compact Riemann surface of genus g . Then:

$$\dim \Omega^1(C) = g$$

Proof :

combine proposition 3.6 and proposition 3.10.

3.2 Two Important Theorems

Let $H_1(C, \mathbb{Z})$ be the collection of closed forms of a genus g Riemann surface. It will have a basis spanned by $2g$ elements. We have a natural dual definition:

Definition 3.12: 1st Cohomology Group

Let C be a compact Riemann surface. Then the first cohomology group of C is:

$$H^1(C, \mathbb{C}) = \text{Hom}_{\mathbb{C}}(H_1(C, \mathbb{Z}), \mathbb{C})$$

Then we saw in proposition 3.7 that we have a map:

$$\begin{aligned} \eta : \{\text{closed differential one-forms on } C\} &\rightarrow H^1(C, \mathbb{C}) \\ \lambda &\mapsto \eta_\lambda \end{aligned}$$

and furthermore this map is injective by proposition 3.8. Let $H_{\text{DR}}(C)$ be the equivalence classes of closed one forms mod exact one forms. Then by proposition 3.8, we have in fact that the above η is a map from $H_{\text{DR}}(C) \hookrightarrow H^1(C, \mathbb{C})$. Then, by the “Hodge” condition, we have that:

$$\begin{aligned} \Omega^1(C) \oplus \overline{\Omega^1(C)} &\hookrightarrow H_{\text{DR}}(C) \\ \omega \oplus \bar{\varphi} &\mapsto \omega + \bar{\varphi} \end{aligned}$$

is also injective. We thus get a triple of injective maps:

$$\Omega^1(C) \oplus \overline{\Omega^1(C)} \hookrightarrow H_{\text{DR}}(C) \hookrightarrow H^1(C, \mathbb{C})$$

Note that the vector spaces on both ends are $2g$, and hence all these spaces are isomorphic! The first isomorphism is a special case of the *Hodge Theorem*, and the second isomorphism is a special case of *DeRham’s Theorem*.

Definition 3.13: Period Vector

Let C be a compact Riemann surface of genus g , $\omega_1, \dots, \omega_g$ a basis for $\Omega^1(C)$, and $\gamma_1, \dots, \gamma_{2g}$ a $\mathbb{Z}$ -basis for $H_1(C, \mathbb{Z})$. Then the $2g$ g -dimensional vector:

$$\pi_i = \begin{pmatrix} \int_{\gamma_i} \omega_1 \\ \vdots \\ \int_{\gamma_i} \omega_g \end{pmatrix}$$

are called a *system of period vectors of C* . The matrix:

$$\Omega = (\pi_1, \dots, \pi_{2g})_{g \times 2g}$$

is called the *period matrix* of C

Proposition 3.14: Period Matrix Nonsingular

The $2g$ period vectors are $\mathbb{R}$ -linearly independent.

Proof :

For the sake of contradiction, suppose $\pi_1, \dots, \pi_{2g}$ are $\mathbb{R}$ -linearly dependent so that there exists $a_1, \dots, a_{2g} \in \mathbb{R}$ not all zero st

$$a_1\pi_1 + \dots + a_{2g}\pi_{2g} = 0$$

We make take the complex conjugate of this (note conjugating an integral brings the conjugation to the inside) and get the matrix:

$$\Omega^* \begin{pmatrix} \pi_1 & \cdots & \pi_{2g} \\ \bar{\pi}_1 & \cdots & \bar{\pi}_{2g} \end{pmatrix}_{2g \times 2g}$$

By our assumption, $\text{Rank}(\Omega^*) < 2g$, hence there is a vector in the kernel

$$\int_{\gamma_i} \left(\sum_{j=1}^g \lambda^j \omega_j + \sum_{j=1}^g \eta^j \bar{\omega}_j \right) = 0 \quad (i = 1, \dots, 2g)$$

Let $\omega = \sum_{j=1}^g \lambda^j \omega_j$ and $\varphi = \sum_{j=1}^g \eta^j \bar{\omega}_j$. Then:

$$\int_{\gamma_i} (\omega + \bar{\varphi}) = 0 \quad (i = 1, \dots, 2g)$$

But then we know $\omega = \varphi = 0$, contradicting the linear independence of $\omega_1, \dots, \omega_g$, as we sought to show.

3.3 Riemann Inequality

Theorem 3.15: Riemann Inequality

Let C be a compact Riemann surface of genus g , $D \in \text{Div}(C)$, and $\deg D = d$. Then;

$$l(D) \geq d - g + 1$$

see [3, p.113] for full details.

Proof :

Like before, let us link Riemann surfaces to curves to get the required context. Let

$$\pi : C \rightarrow C' \subseteq \mathbb{CP}^2$$

be the normalization map with δ ordinary double points $p_1, \dots, p_\delta$. Griffith's offers the following visualization:

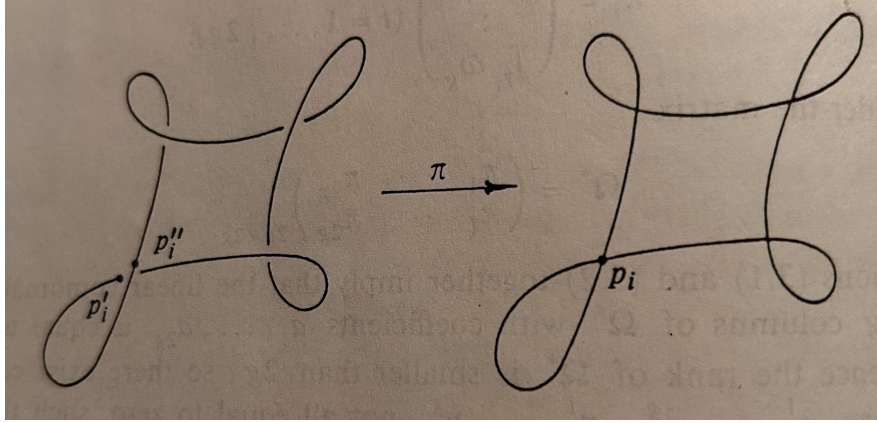


Figure 1: normalizatoion

Let $\pi^{-1}(p_i) = p'_i + p''_i$. Then let:

$$\Delta = \sum_{i=1}^{\delta} (p'_i + p''_i) \in \text{Div}(C)$$

Then by a change of coordinates in $\mathbb{CP}^2$, without loss of generality we may assume neither $\pi(D)$ nor $\pi(\Delta)$ contain the point at infinity (like in the above figure). Now, let C' be given by the equation $F(\xi^0, \xi^1, \xi^2) = 0$, $\deg F = m$. Let $D = D' - D''$ where $D', D'' \geq 0$ and let

$$\deg D' = d' \quad \deg D'' = d''$$

Let S^n be the vector-space of complex homogeneous polynomials in three variables of degree n . Then:

$$\dim S^n = \frac{1}{2}(n+1)(n+2)$$

Now let's consider a polynomial $G(\xi^0, \xi^1, \xi^2) \in S^n$ satisfying:

1. $F \nmid G$
2. $G \cdot C \geq D' + \Delta$ where $G \cdot C = (\pi^*(G(\xi^0, \xi^1, \xi^2)))$ (that is, the divisor induced by the zeros of $\pi^*(G)$ on C). Note that $G \cdot C \in \text{Div}(C)$.

We shall elaborate why we would want such a polynomial, then show such a polynomial exists. First, for $p \in C$, let $p' = \pi(p) \in C'$. Then a neighbourhood of p gets mapped under π into a curve component of C' near p' ; denote it by B_p . Then after a possible coordinate change, we may take p' to be $[1 : 0 : 0]$. Label the equation after this coordinate change G_p . As the multiple points of C' are ordinary double points, the equation of B_p near p' is:

$$y = a_0 + a_1 x^2 + \dots$$

Putting this into $G_p(1, x, y)$, we get a power series in terms of x . Now, let $n(p)$ denote the degree of the lowest term. We then get:

$$G \cdot C = \sum_{p \in C} n(p)p$$

the right hand side is certainly a finite sum. (I'll finish rest later)

Corollary 3.16: Comparing Degree and Genus

If $d \geq g$, then

$$l(D) \neq 0$$

Hence if $l(D) = 0$, then:

$$d \leq g - 1$$

Proof :

immediate

Proposition 3.17: Meromorphic Differentials and Riemann Inequality

$$i(D) \geq g - d - 1$$

Proof :

Let F, m, D', D'', d', d'' , and Δ be as in theorem 3.15. **finish here.**

Corollary 3.18: Comparing Degree and Genus – Meromorphic Differential

If $d \leq g - 2$, then $i(D) \neq 0$, hence if $i(D) = 0$ then

$$d \geq g - 1$$

Proof :

immediate

Finally this leads us to:

Theorem 3.19: Riemann-Roch Theorem

Let C be a compact Riemann surface of genus g and :

$$D = \sum_{i=1}^k n_i p_i \in \text{Div}(C)$$

Then:

$$l(D) = d - g + i(D) + 1$$

Proof :

see [3, p.118]

Let us first assume $D \geq 0$ and prove

$$l(D) \leq d - g + i(D) + 1$$

And then take cases and judiciously use the Riemann inequality and its corollary to get the results. Suppose $\eta = (\eta_1, \dots, \eta_k)$ where:

$$\eta_i = \frac{a^{in_i}}{z_i^{n_i}} + \dots + \frac{a_{i1}}{z_i}$$

is a Laurent principal part near p_i . The linear space consisting of all these η is isomorphic to $\mathbb{C}^d$. Then thinking of η as a vector in $\mathbb{C}^d$, consider:

$$\begin{aligned} \text{Res} : \mathbb{C}^d \times \Omega^1(C) &\rightarrow \mathbb{C} \\ (\eta, \omega) &\mapsto \sum_{i=1}^k \text{Res}_{p_i}(\eta_i \omega) \end{aligned}$$

This is certainly a bilinear mapping. Let us show that:

$$\text{Res}(\eta, \omega) = 0, \quad \forall \eta \in \mathbb{C}^d \quad \text{if and only if} \quad \omega \in \Omega^1(D)$$

If $\omega \in \Omega^1(D)$, then $\eta_i \omega$ does not have p_i as a pole for all $i = 1, \dots, k$, hence:

$$\text{Res}_{p_i}(\eta_i \omega) = 0$$

and so the sum of residues is zero, giving the $\Leftarrow$ direction. For the $\Rightarrow$ direction, assume otherwise so that $\omega \in \Omega^1(C)$ such that $\text{Res}(\eta, \omega) = 0$ for all $\eta \in \mathbb{C}^d$ but $\omega \notin \Omega^1(D)$. Then near p_i , let

$$\omega = b_{i0} z_i^{m_i} + b_{i1} z_i^{m_i+1} + \dots$$

where $m_i \in \mathbb{Z}$, $b_{i0} \neq 0$. By assumption, $m_i < n_i$. Then there is a Laurent Principal part near p_i

$$\xi_i = (\xi_{i1}, \dots, \xi_{ik})$$

such that

$$\xi_{ii} = \frac{1}{z_i^{m_i+1}} \quad \xi_{ij} = 0 \quad (i \neq j)$$

As $m_i + 1 \leq n_i$, $\mathbb{C}^d$, by treating these as $\mathbb{C}^d$ we have $\xi_i \in \mathbb{C}^d$. However:

$$0 = \text{Res}(\xi_i, \omega) = \text{Res}_{p_i} \frac{b_{i0}}{z_i} = b_{i0} \neq 0$$

a contradiction. Thus, $m_i \geq n_i$, and so $\omega \in \Omega^1(D)$.

Next, recall the following fact (or see exercise ref:HERE). Let $B : V \times W \rightarrow \mathbb{C}$ be a bilinear map between two finite dimensional vector spaces, in particular $\dim V = n$ and $\dim W = m$.

Then B is called non-degenerated if given $w \in W$, $B(v, w) = 0$ for all $v \in V$ implies $w = 0$. Then given $L \subseteq V$ satisfies $B(L \times W) = 0$, then:

$$\dim L \leq n - m$$

Then by the property we just found earlier about the residue map, we have a non-degenerate bilinear map:

$$\text{Res}_1 : \mathbb{C}^d \times \frac{\Omega^1(C)}{\Omega^1(D)} \rightarrow \mathbb{C}$$

Now recall we can interpret $\mathcal{L}(D)/\mathbb{C}$ as a subspace of $\mathbb{C}^d$, and by what we've just shown:

$$\text{Res}_1 \left(\frac{\mathcal{L}(D)}{\mathbb{C}} \times \frac{\Omega^1(C)}{\Omega^1(D)} \right) = 0$$

Then:

$$\dim \left(\frac{\mathcal{L}(D)}{\mathbb{C}} \right) \leq \dim \mathbb{C}^d - \dim \left(\frac{\Omega^1(C)}{\Omega^1(D)} \right)$$

That is:

$$l(D) - 1 \leq d - \dim \Omega^1(C) + i(D)$$

By theorem 3.5, $\dim \Omega^1(C) = g$, and so we get:

$$l(D) \leq d - g + i(D) + 1$$

This was the hardest part of the proof. We now cases cases on $l(D)$ and $i(D)$ to finish:

1. Consider $l(D) \neq 0$, $i(D) = 0$. As $l(D) \neq 0$, there exists an $f \in K(C)$ such that $(f) + D \geq 0$. Let $E = (f) + D$. Then $(f) = E - D$, so $E \sim D$. Then

$$l(D) = l(E) \quad i(D) = i(E) \quad \deg D = \deg E$$

Thus, we may assume $D \geq 0$. Thus we get our inequality, and by the Riemann inequality:

$$d - g + i(D) + 1 = d - g + 1 \stackrel{\text{R.}}{\leq} l(D) \leq d - g + 1$$

giving equality

2. Consider $l(D) \neq 0$, $i(D) \neq 0$. As $l(D) \neq 0$, we may do the same trick to assume $D \geq 0$. Since $i(D) \neq 0$, there exists a $\omega \in \Omega^1(C)$ such that

$$(\omega) \geq D$$

Let $E = (\omega) - D$ so that $E \geq 0$. Then by proposition 3.3 and proposition 3.17 we get:

$$i(D) = l(E) \leq \deg E - g + 1 + i(E)$$

By the Poincaré-Hopf formula, $\deg(\omega) = 2g - 2$ so

$$\deg E = \deg(\omega) - \deg(D) = 2g - 2 - d$$

As $i(E) = l(D)$, we get:

$$i(D) \leq 2g - 2 - d - g + 1 + l(D)$$

re-arranging and combining we get:

$$l(D) \geq d - g + i(D) + 1$$

giving us the other wise

3. Consider $l(D) = 0$, $i(D) \neq 0$. Since $i(D) \neq 0$, there exists a $\omega \in \Omega^1(C)$ such that $(\omega) = D + E$ where $E \geq 0$. Then by proposition 3.3

$$l(E) = i(D) \neq 0 \quad i(E) = l(D) = 0$$

Thus, by case (1) applied to E we get:

$$l(E) = \deg E - g + i(E) + 1$$

Note that

$$\deg E = \deg(\omega) - \deg D = 2g - 2 - d$$

Then substituting and rewriting $l(E)$, $i(E)$ as $i(D)$ and $l(D)$, we get the equality.

4. Consider $l(D) = 0$, $i(D) = 0$. Then by corollary 3.16 and corollary 3.18

$$d = g - 1$$

which is the desired equation.

(then do the exercises for $C = \mathbb{P}^1$ direct prove, and the elliptic curve exercise)

4 Application of Riemann-Roch Theorem

It shall cover genus 0, 1, 2, 3, 4.

4.1 Genus 0

Proposition 4.1: Genus 0 then Rational

Let C be a compact Riemann surface of genus 0. Then

$$C \cong \mathbb{P}^1$$

Proof :

Let $p \in C$, $D = (p) \in \text{Div}(C)$. Then:

$$0 \leq i(D) \leq i(0) = \dim \Omega^1(C) = 0$$

Thus $i(D) = 0$. By the Riemann-Roch theorem:

$$l(D) = d - g + i(D) + 1 = 1 - 0 + 0 + 1 = 2$$

Thus, $\mathcal{L}(p)$ contains two linearly independent function, one of which can be chosen to be $f_1 = 1$. The other function must be a meromorphic function (as C is a compact Riemann surface) and must have p as a pole of order 1; call this function f_2 . Then by definition $f_2^{-1}(\infty) = p$. Hence it follows that this is a holomorphic mapping:

$$f_2 : C \rightarrow \mathbb{P}^1$$

where $\deg f_2 = \#\{f_2^{-1}(\infty)\} = 1$ implying that f_2 is in fact biholomorphic (recall lemma 2.2). Thus:

$$C \cong \mathbb{P}^1$$

as we sought to show.

4.2 Genus 1

Proposition 4.2: Genus 1 Weierstrass Equation Representation

Let C be a compact Riemann surface of genus 1. Then C can be represented by a smooth algebraic curve of degree 3 in $\mathbb{P}^3$

Proof :

We shall construct a holomorphic injective map $f : C \hookrightarrow \mathbb{P}^2$ where f shall be a smooth algebraic curve of degree 3. First, for any nonzero $\omega \in \Omega^1(C)$, by Poincaré-Hopf formula we have:

$$\deg(\omega) = 2g - 2 = 2 - 2 = 0$$

Hence, if $\omega(p) = 0$, then it would have to have a pole, but it is holomorphic! Thus, for any $D > 0$, $i(D) = 0$. Thus, letting $D = 2p$, we get:

$$i(D) = 0$$

Thus by the Riemann-Roch theorem we have that:

$$l(2p) = d - g + i(2p) + 1 = 2 - 1 + 0 + 1 = 2$$

Hence $\mathcal{L}(D)$ has non-constant meromorphic functions which only have p as a singular point of order ≤ 2 . In fact, it *must* be of order 2, as by the Riemann-Roch theorem:

$$l(p) = 1 - 1 + 1 = 1$$

Hence $\mathcal{L}(p)$ only has constant functions; and so the order of any meromorphic function with single pole p is at least 2. Let $x \in \mathcal{L}(2p)$ be such a function. Then near p we have:

$$x = \frac{c_2}{t^2} + \frac{c_1}{t} + h_1(t)$$

for a local coordinate near p where $t(p) = 0$ and $c_2 \neq 0$. Let's start by showing that it must be that $c_1 = 0$. As $\dim \Omega^1(C) = g = 1$, take a $\omega \in \Omega^1(C)$ that is nonzero so that under t

$$\omega = g(t)dt \quad g(0) \neq 0$$

Then $x\omega$ has no poles other than p , so by the Residue Theorem (the sum of all residues is 0):

$$0 = \text{Res}_p(x\omega) = c_1$$

Thus, with a suitable change of coordinates we get:

$$x = \frac{1}{z^2} + h(z)$$

Note that this x has many similar properties to the Weierstrass function, as we shall see later.

Now, for any $\omega \in \Omega^1(C)$, let

$$y = \frac{dx}{\omega}$$

which is well defined as $\omega(p) \neq 0$ for all $p \in C$. Define:

$$\begin{aligned} f : C &\rightarrow \mathbb{P}^2 \\ q &\mapsto [1 : x(q) : y(q)] \quad (p \mapsto [0 : 0 : 1]) \end{aligned}$$

Note that our chosen point p was mapped to the point at infinity. Certainly f is holomorphic, we must just show it is injective and that $f(C)$ is a curve of degree 3.

Starting by showing $f(C)$ is a degree 3 curve in $\mathbb{P}^2$, consider $x : C \rightarrow \mathbb{P}^1$. Then by construction $x^{-1}(\infty) = 2p$. Thus:

$$\deg x = 2$$

Thus, the ramification can never exceed 1, and hence the ramification index at its ramification points never exceeds 1. In particular, the coefficients of every point for the ramification divisor of x must be 1. By the Riemann-Hurwitz formula;

$$\deg R = 2(\deg x + g - 1) = 2(2 + 1 - 1) = 4$$

As p ramified, there are 3 other Ramification points:

$$R = p_1 + p_2 + P_3 + p$$

each p_i being mutually distinct:

FIG HERE

let $x(p_i) = a_i$. Let us figure out what type of ramification we have. To be succinct with tracking roots and poles, for a meromorphic differential form $\alpha \in K^1(C)$, let $(\alpha)_0$ and $(\alpha)_\infty$ represented the zeros and poles (including multiplicity) respectively. As ω has no zeros or poles then since $y = dx/\omega$ we get:

$$(y)_0 = (dx)_0 \quad (y)_\infty = (x)_\infty$$

Hence, $(y)_0$ and $(y)_\infty$ can only contain the points of x with ramification index 1, that is p_1, p_2, P_3, p . Thus, each of these “zero” and “pole” divisors are shared between the two.

To calculate, suppose $z_i (i = 1, 2, 3)$ are local coordinates near p_i where $z_i(p_i) = 0$. Since the multiplicity of x at p_i is 2,

$$x - a_i = z_i^2 h_i(z_i)$$

where h_i is a holomorphic function and $h_i(0) \neq 0$. Thus:

$$\begin{aligned} dx &= 2z_i h_i(z_i) dz_i + z_i^2 h_i'(z_i) dz_i \\ &= z_i (2h_i(z_i) + z_i h_i'(z_i)) dz_i \end{aligned}$$

But then evaluating at $z_i = 0$:

$$(2h_i(z_i) + z_i h_i'(z_i))|_{z_i=0} = 2h_i(0) \neq 0$$

and so p_i is a simple zero of dx . For the point p , choosing a local coordinate function z near p as we did before we get:

$$x = \frac{1}{z^2} + h(z)$$

Neurally, dx has p as a pole of order 3, and so we found:

$$(y)_0 = p_1 = P_2 + P_3 \quad (y)_\infty = 3p \quad (y) = p_1 + p_2 + p_3 - 3p$$

We now know the roots/pole information our desired function must have. Let $g(x) \in K(\mathbb{P}^1)$ be given by:

$$g(x) = (x - a_1)(x - a_2)(x - a_3)$$

Considering g as a function on C (that is x^*g), then as near the points p_i and p in coordinates z_i and z :

$$x - a_i = z_i^2 h_i(z_i) \quad x = \frac{1}{z^2} h(z)$$

we get:

$$(g)_0 = 2p_1 + 2p_2 + 2p_3 \quad (g)_\infty = 6p$$

since each p_i takes p_i as a zero of order 2, and near p every $(x - a_i)$ takes p as pole of order 2.

Now, we see that the divisor of g is just the divisor of y^2 , and so:

$$\left(\frac{y^2}{g} \right) = 0$$

showing y^2/g is holomorphic on C , and as C is compact $y^2/g = a \in \mathbb{C}$ where $a \neq 0$. To choose a better constant replace y by $y/\sqrt{a}$. Then we get:

$$y^2 = g(x)$$

But now, we just showed that under f that C lies in the image of a curve of degree 3:

$$C' = \{[1 : x : y] \in \mathbb{P}^2 \mid y^2 - g(x) = 0\} \cup \{[0 : 0 : 1]\}$$

Giving us the degree. Let us now check that f is an injective mapping except at ∞ .

To that end, we shall need a bit more structure on C . We shall define a natural involution map on C . Let

$$j : C \rightarrow C \\ q_1 \mapsto q_2$$

where $q_1, q_2 = x^{-1}(p_r)$ are an arbitrary $p_r \in C'$ (where I chose subscript r for no particular reason, we're running out of letters). Then certainly $j^2 = \text{id}$. It is thus its own inverse, and so it is easy to verify it is an automorphism. Using this, we shall show some properties of j^* on $K(C)$, $K^1(C)$ and $\Omega^1(C)$. Certainly by definition of j :

$$j^*(x) = x \quad j^*(dx) = dx$$

Let's show $j^*y = -y$. As $y = dx/\omega$:

$$j^*y = \frac{j^*(dx)}{j^*(\omega)} = \frac{dx}{d^*(\omega)}$$

Hence, we show $j^*(\omega) = -\omega$. As $\dim \Omega^1(C) = g = 1$, we get $\Omega^1(C) \cong \mathbb{C}$. Then notice that j can be thought of as a linear transformation on $\Omega^1(C)$ so that $j^2 = 1$ implies $(j^*)^2 = 1$, (hence it is diagonalizable) hence the eigenvalues of j^* are either 1 or -1 . Suppose j^* has eigenvalue 1 so that $j^*\omega = \omega$. Then there exists a holomorphic differential φ on $\mathbb{P}^1$ such that

$$\omega = \varphi \circ x = x^* \varphi$$

(exercise). But there is no nonzero holomorphic differential on $\mathbb{P}^1$, hence $j^*\omega = -\omega$.

With the properties of j established, consider $q_1, q_2 \in C$ where $f(q_1) = f(q_2)$. We want to show that $q_1 = q_2$. Naturally if either q_1, q_2 is p then by construction $f(q_1) = f(q_2) = \infty$ and so $q_1 = q_2 = p$; thus we assume neither q_1, q_2 is p . By definition:

$$\begin{aligned} f(q_1) &= [1 : x(q_1) : y(q_1)] \\ f(q_2) &= [2 : x(q_2) : y(q_2)] \end{aligned}$$

and as these are equal $x(q_1) = x(q_2)$ and $y(q_1) = y(q_2)$. Now if $q_1 \neq q_2$, as $x(q_1) = x(q_2)$ it must be that $j(q_1) = q_2$. Thus, we have by definition of the pullback:

$$j^*y(q_1) = y(q_2)$$

however as $j^*(y) = -y$

$$j^*y(q_1) = -y(q_1) = -y(q_2)$$

and so

$$y(q_2) = -y(q_2)$$

telling us $y(q_2) = 0$. But $(y)_0 = p_1 + p_2 + p_3$, so $q_2 \in \{p_1, p_2, p_3\}$. Thus:

$$q_1 = j(q_2) = q_2$$

but this contradicts $q_1 \neq q_2$, thus it must be that they are equal, showing f is injective.

Let's now show surjectivity. We shall use a connectedness argument. First, $y^2 - g(x) = 0$ is certainly an irreducible polynomial, hence it forms an irreducible subset, hence a connected subset. Next, C is compact, and as f is holomorphic $f(C)$ is closed. As f is holomorphic, $f(C)$ is also open (by the holomorphic open mapping theorem), and so $f(C) = C'$.

Finally, we prove C' is smooth.

Index

1st Cohomology Group, 17

Laurent Principal Part, 9

Linearly Equivalent, 8

Normalization, 4

Period Vector, 17

Ramification Divisor, 5

References

- [1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. 1 vols. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.
- [2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Complex Analysis*. 1st ed., p. 233. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_complex_analysis.pdf.
- [3] Phillip A. Griffiths. *Introduction to Algebraic Curves*. Vol. 78. Mathematical Monographs. American Mathematical Society.